



Effective and group refractive index extraction and cross-sectional dimension estimation for silicon-on-insulator rib waveguides

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Abstract: The accurate determination of the effective and group refractive indices (n_{eff} and n_g) of optical waveguides as a function of wavelength is of critical importance to the design of photonic integrated circuits (PICs). This paper demonstrates the extraction of the two parameters of silicon-on-insulator (SOI) rib waveguides using the transmission spectra of two racetrack micro-ring resonators (MRRs) with different perimeters. The extracted n_{eff} and n_g exhibit an uncertainty of approximately 10^{-3} . Based on the extracted $n_{\text{eff}}(\lambda)$, we estimate the cross-sectional dimension of the SOI rib waveguide that constitutes the MRR. This waveguide has a nominal rectangular cross section with a width, height, and slab thickness of 450 nm, 200 nm, and 70 nm, respectively. The estimated cross-sectional dimension is in accordance with the findings of the scanning transmission electron microscopy (STEM) analysis, exhibiting a discrepancy of approximately 1%. The proposed methodology offers a universal approach to n_{eff} and n_g extraction and a non-invasive method for cross-sectional dimension assessment, which can be applied in different PIC platforms.

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1. Introduction

Silicon photonics has the potential to facilitate the miniaturization of photonic devices due to the high refractive index contrast between silicon and its native oxide. The utilization of complementary metal-oxide-semiconductor (CMOS) compatible manufacturing technologies has enabled considerable advancements in the design, fabrication and applications of silicon photonics [1,2]. A plethora of components have been developed for use in a multitude of applications on silicon photonic platforms. These include waveguides, ring resonators, couplers, modulators, switches, wavelength division multiplexers, and photodetectors [3–10]. Nevertheless, the susceptibility of silicon PIC device performance to fabrication imperfections represents a significant obstacle in the design of silicon PICs [11–13]. This is, in fact, a consequence of the high refractive index contrast inherent in the material and the resulting miniaturization of the structure.

In the design and fabrication of silicon photonic devices, a notable discrepancy is often observed between the actual performance of the fabricated devices and the simulation results predicted during the design phase, even when the devices have undergone meticulous simulation [11]. The discrepancy between the ideal design and the actual fabricated silicon photonic devices is mainly due to uncontrollable factors in the fabrication process. These factors cause deviations in the size, shape, and material properties of the fabricated devices, which in turn affect their performance [14].

Consequently, it is crucial to ascertain the practical cross-sectional dimensions of the fabricated devices across the wafer for yield evaluation. A plethora of methodologies have been proposed in the literature for this purpose. The most straightforward approach is to dice the photonic chip and

assess the dimensions of the waveguides using a scanning electron microscope (SEM). However, this approach is costly, time-consuming, and causes permanent damage to the chip. Furthermore, it is not suitable for wafer-scale evaluation. Another way to get geometric parameters is through the analysis of the response of complex structures, including MRRs [11,14–17] and Mach-Zehnder interferometers (MZIs) [12,18–23]. Others rely on complex reflection spectral measurements in Bragg gratings [24,25], waveguide junctions [26], and point-reflector optical waveguides [27]. The MRR and MZI structures require relatively simple measurements, with much work devoted to extracting the effective refractive index, group refractive index, and waveguide cross section [11–23]. Nevertheless, the attainment of sufficiently precise outcomes frequently necessitates the presence of specific conditions, such as a minimal degree of process error and the utilization of a multitude of test structures. Moreover, the majority of the demonstrated methods extract only the width and height data of strip waveguides (i.e., channel waveguides) [12,14,19,25]. However, the actual cross-sectional shape of the waveguide is typically trapezoidal. It means that the sidewall angle is also a critical dimensional parameter. Moreover, it is frequently necessary to extract structural parameters for rib waveguides, which is of critical importance for the design of silicon electro-optic modulators. The evaluation method for rib waveguides with a trapezoidal cross section (i.e., four structural parameters of width, height, slab thickness, sidewall angle) has not yet been investigated.

To address this issue, we propose a methodology for the analysis of the spectral responses of two racetrack MRRs in order to determine the effective and group refractive index functions ($n_{\text{eff}}(\lambda)$ and $n_g(\lambda)$) of the waveguide that constitutes the MRR. The process is illustrated in *Step One* of Fig. 1. The function $n_{\text{eff}}(\lambda)$ is determined by a search algorithm, where the figure of merit (FOM) is defined as the difference between the calculated and measured resonance locations. Subsequently, the extracted $n_{\text{eff}}(\lambda)$ is employed to ascertain the cross-sectional dimension of the waveguide, as illustrated in *Step Two* of Fig. 1. The geometric parameters of this trapezoidal waveguide are determined through an algorithm analogous to that employed in the preceding step, wherein the FOM is defined as the difference between the simulated $n_{\text{eff}}(\lambda)$ and the extracted function obtained in *Step One*. Since MRR is sensitive to temperature, a thermoelectric controller (TEC) was used to maintain the temperature at 25°C throughout the experiment to ensure accurate acquisition of the micro-ring spectra. This method prevents any potential drift in the resonance peak positions. Simulations were also performed at 25 °C.

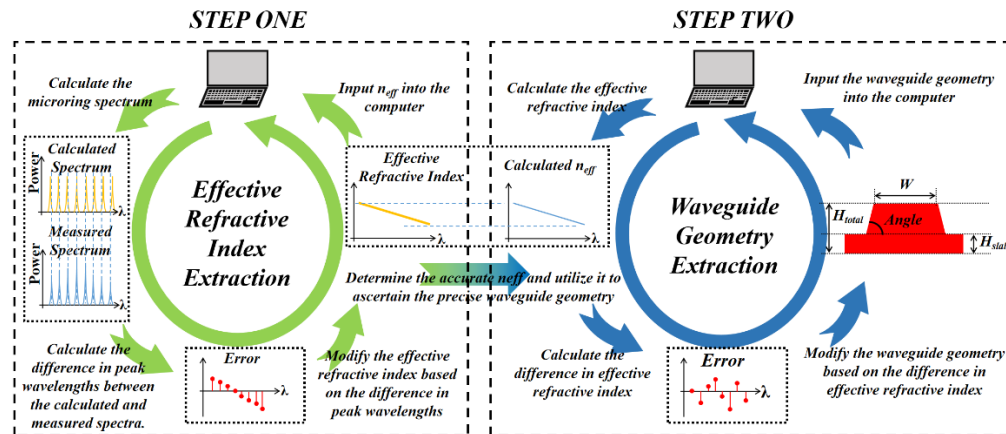


Fig. 1. The procedure for extracting the effective refractive index (*Step One*) and determining the waveguide cross section geometry (*Step Two*).

A comparison of the extracted result with the waveguide cross section measured with STEM revealed a discrepancy of approximately 5 nm and 2 nm in width and height, respectively. This suggests that the estimated dimensional error is approximately 1%. This method offers a simple and effective approach to extract the $n_{\text{eff}}(\lambda)$. Additionally, it offers a non-invasive approach to estimating the cross-sectional dimensions of the waveguide. The method can be applied to a wide range of material systems and structures, including indium phosphide (InP) and lithium niobate on insulator (LNOI), thus making it a versatile tool for waveguide characterization and fabrication process evaluation.

The paper is organized as follows. In section 2, we demonstrate, through theoretical analysis, that the effective refractive index of a waveguide can be accurately determined in the form of a cubic polynomial from the spectra of two racetrack MRRs with the same cross section and different perimeters. In section 3, we introduce an improved particle swarm optimization (PSO) algorithm for the purpose of identifying the optimal refractive index, denoted by $n_{\text{eff}}(\lambda)$, in the form of a cubic polynomial that produces transmission spectra most closely aligned with the measured spectra of the two MRRs. In section 4, the same algorithm is employed to ascertain the cross section of the silicon rib waveguide. The optimal structural parameters, including width, height, slab thickness, and sidewall angle, are identified to produce an effective refractive index that matches the $n_{\text{eff}}(\lambda)$ function determined in the previous section. To demonstrate the accuracy of our results, we conducted a simulation of the n_{eff} and n_g of the waveguide cross section obtained by high-resolution STEM and compared the results with our extracted values. The maximum errors observed are 0.0014 for the n_{eff} and 0.0018 for the n_g . The discrepancy between the extracted and measured dimensions is approximately 5 nm in width and 2 nm in height. Finally, we evaluate our method in comparison to the current state of the art in non-invasive waveguide characterization. The method employed herein is capable of extracting the maximum number of waveguide geometry parameters.

2. Theoretical analysis for the numerical fitting of n_{eff}

The objective is to obtain the waveguide cross section through a straightforward spectral measurement process, without resorting to chip dicing. To achieve this objective, we select the MRR for spectral measurement because of its compact footprint and dense, high-finesse spectral response. Once the layout of the device is determined, the resonance wavelength locations of the MRR are solely dependent on the effective refractive index of the waveguide (i.e., the $n_{\text{eff}}(\lambda)$ function).

We will show in simulations that the $n_{\text{eff}}(\lambda)$ function can be accurately represented by a cubic polynomial. The cross-sectional dimension of a silicon rib waveguide is defined by four variables: width (W), height (H), slab thickness (T), and sidewall angle (A). In the finite difference method (FDM) simulations and mean squared error (MSE) calculations in Fig. 2, we adopt the following values for W , H , T , and A : 450 ± 50 nm, 200 ± 10 nm, 70 ± 10 nm, and $80 \pm 10^\circ$, respectively. Given that each parameter has three values, there are 3^4 possible combinations, resulting in a total of 81 distinct cross sections, as indicated by the structure number in Fig. 2. In addition, we use 10 wavelengths equally distributed between 1525 nm and 1565 nm to calculate the MSE. The definition of MSE in this paper is as follows.

$$MSE = \frac{1}{N} \sum_{i=1}^N [n_{\text{eff}}^{\text{fit}}(\lambda_i) - n_{\text{eff}}^{\text{sim}}(\lambda_i)]^2 \quad (1)$$

where $n_{\text{eff}}^{\text{fit}}$ is the effective refractive index obtained by polynomial fitting, $n_{\text{eff}}^{\text{sim}}$ is the simulated effective refractive index, and N is the number of wavelengths, which in this instance is 10.

As illustrated in Fig. 2, for all the 81 cross sections, the MSE of the polynomial fitting converges when the order of the polynomial is greater than three. Accordingly, a cubic polynomial is utilized

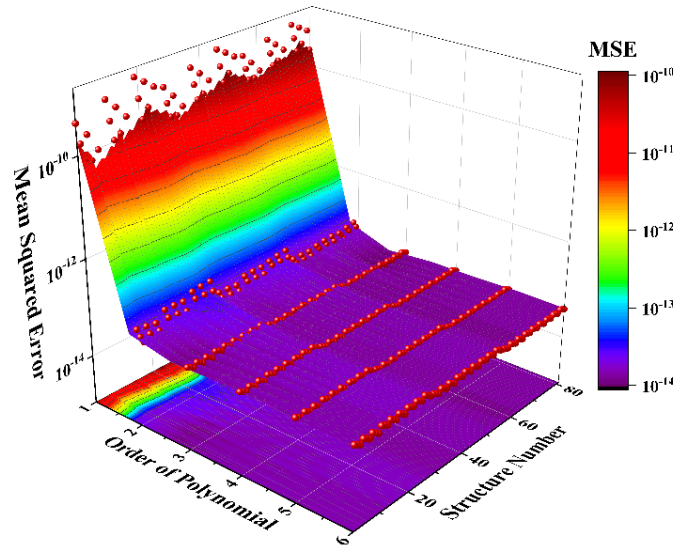


Fig. 2. The MSE associated with polynomial fitting of varying orders for a range of silicon rib waveguide cross sections.

to represent the effective refractive index as a function of wavelength, as illustrated below:

$$n_{\text{eff}}(\lambda) = A \cdot \lambda^3 + B \cdot \lambda^2 + C \cdot \lambda + D \quad (2)$$

The resonance condition of the MRR is:

$$n_{\text{eff}}(\lambda) \cdot L = m \cdot \lambda \quad (3)$$

In Eq. (3), the variable L represents the perimeter of the cavity of the MRR, while the variable m is an integer, specifically the resonance order, which varies with different resonance wavelengths. Upon substituting Eq. (3) into Eq. (2), the following result is obtained.

$$A \cdot \lambda^2 + B \cdot \lambda + C + \frac{D}{\lambda} = \frac{m}{L} \quad (4)$$

For each resonance wavelength, it should satisfy Eq. (4) with a specific resonance order m . Let us suppose that we have algorithmically determined a cubic polynomial $n_{\text{eff}}(\lambda)$, that is, we have found a set (A, B, C, D) that produces a series of resonance wavelengths that best match the measured spectra. In this case, we can add a constant real number Q to both the left and right sides of Eq. (4) as follows:

$$A \cdot \lambda^2 + B \cdot \lambda + C + \frac{D}{\lambda} + Q = \frac{m + Q \cdot L}{L} \quad (5)$$

When QL is an integer, which can be achieved by ensuring that Q is a constant real number with an arbitrary value, the above equation can be expressed as follows:

$$A \cdot \lambda^2 + B \cdot \lambda + C' + \frac{D}{\lambda} = \frac{m'}{L} \quad (6)$$

This indicates that the presence of an additional group (A, B, C', D) , which represents another $n_{\text{eff}}(\lambda)$ function, also ensures that the calculated resonance wavelengths are aligned with the

measured resonance wavelengths with the same degree of precision. Consequently, an infinite number of $n_{\text{eff}}(\lambda)$ can be obtained with a single MRR. In other words, it is not possible to determine the resonance order m and the linear coefficient C in the cubic polynomial with certainty using a single MRR.

In order to address this issue, we propose the introduction of an additional MRR with the identical cross-section, identical bend radius and a different circumference (L'). Given that the two MRRs share the identical cross section, they have the same nominal $n_{\text{eff}}(\lambda)$, that is, the same set of (A, B, C, D) . In order for a new solution to be viable, it is necessary that both QL and QL' must be integers simultaneously. Consequently, the following equations will exist:

$$Q = \frac{K}{L} = \frac{K'}{L'} \quad (7)$$

The symbols K and K' represent two distinct integers. In the case of two racetrack MRRs comprising both straight and circular waveguides, the two circumferences, L and L' , can be expressed as follows:

$$L = 2L_{\text{Straight}} + 2\pi R, L' = 2L'_{\text{Straight}} + 2\pi R'. \quad (8)$$

We incorporate Eq. (8) into Eq. (7) to derive the subsequent equation:

$$K' \cdot L_{\text{Straight}} - K \cdot L'_{\text{Straight}} = \pi \cdot (K \cdot R' - K' \cdot R) \quad (9)$$

The left side of Eq. (9) is rational or even integer if integer multiples of microns are chosen for the two straight waveguides (i.e., the L_{Straight} and L'_{Straight}). Since π is irrational in the right side of Eq. (9), it is impossible for this equation to hold if we also choose integer multiples of microns for the two radii (i.e., the R and R'). This implies that Eq. (7) is also invalid. To put it another way, it is not possible to add a constant real number Q to the fitted cubic polynomial with coefficients (A, B, C, D) in order to produce another polynomial that satisfies the resonance conditions for both MRRs. Consequently, the existence of two MRRs with identical cross-sectional dimensions but differing perimeters ensures the existence of a unique solution for the effective refractive index function fitting.

The above calculations are based on the assumption that the straight waveguide, the curved waveguide and the coupling regions have the same effective refractive index. However, for a waveguide bend with a small radius, the effective refractive index may differ significantly from that of a straight waveguide. The effective refractive index for curved waveguides with different radii is calculated using the FDM, as illustrated in Fig. 3(a). In the simulations, we assume a silicon rib waveguide with dimensions of $450 \text{ nm} \times 200 \text{ nm}$, a slab thickness of 70 nm , and a sidewall angle of 90° . In Fig. 3(b), the MSE between the n_{eff} of the curved waveguide and that of the straight waveguide with 10 wavelength samples is calculated. As shown in Fig. 3(b), the n_{eff} difference between the straight and bent waveguides is almost negligible when the radius exceeds $20 \mu\text{m}$. Therefore, we adopt a radius of $20 \mu\text{m}$ in our device design. It should be noted that this condition only applies to SOI waveguides or waveguides with a larger refractive index contrast. In the case of silicon nitride waveguides or other waveguides with a low refractive index contrast, a larger radius is required.

We have showed that the n_{eff} of curved and straight waveguides are almost identical when the radius is greater than $20 \mu\text{m}$. In the following, we will discuss the effect of an adjacent straight waveguide on the n_{eff} of a curved waveguide. We will show that the effect of an adjacent straight waveguide is negligible when the gap spacing between them is greater than 500 nm , which is exactly the design in our devices.

As shown in Eq. (10), the phase shift of light after propagating in a waveguide is determined by the n_{eff} of the waveguide, the propagation distance (L) and the wavelength in vacuum (λ).

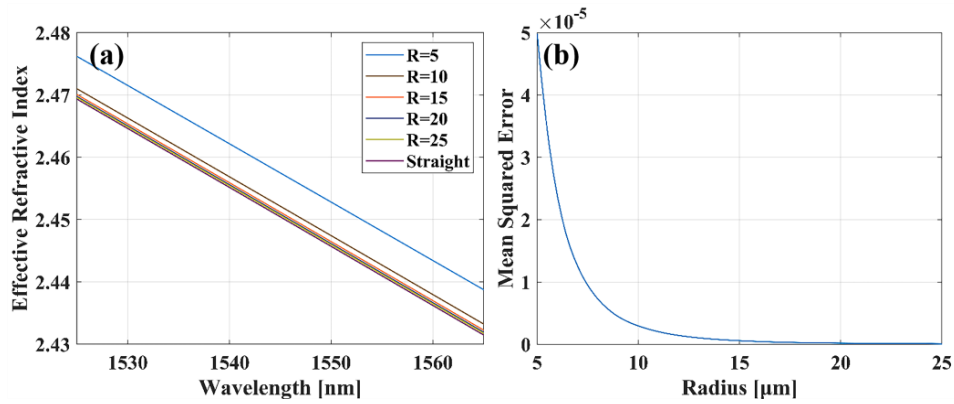


Fig. 3. (a) Effective refractive index of straight waveguide and bent waveguide with different radius. The bottom curve is the n_{eff} of the straight waveguide. (b) MSE of the effective refractive index for 10 equally spaced wavelengths in the C-band between a straight waveguide and a bent waveguide with different radii.

Therefore, we can obtain the n_{eff} of the two structures shown in Fig. 4 by measuring the phase shifts with numerical simulations. The effect of the coupling region on the effective refractive index can be obtained by comparing the n_{eff} of the two structures in Figs. 4(a) and (b).

$$\phi = 2\pi n_{\text{eff}} \frac{L}{\lambda} \quad (10)$$

We input a continuous wave of different wavelength from the input port shown in Fig. 4 and measure the phase difference between the optical electric fields at the output and input ports. We can obtain the n_{eff} at different wavelengths using Eq. (10). The simulation results are shown in Fig. 5. It can be seen from Fig. 5(a) that the influence of the adjacent straight waveguide on the n_{eff} of a 20 μm radius curved waveguide is negligible when the gap spacing is large enough. Figure 5(b) shows the MSE of the n_{eff} for 10 equally spaced wavelengths in the C-band between the two structures in Fig. 4 with different gap spacings. In our device design, we choose a gap spacing of 500 nm, which has a negligible effect on the n_{eff} of the 20 μm radius curved waveguide as shown in Fig. 5. Such a large gap spacing can produce a high-finesse spectral response, allowing more accurate determination of the resonance wavelength.

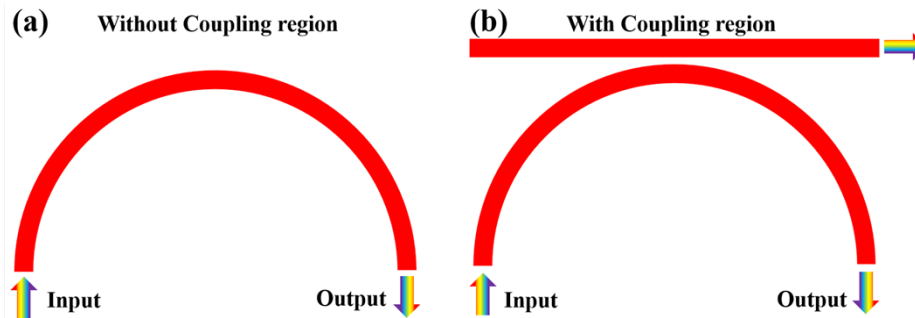


Fig. 4. (a) Bend waveguide without coupling region. (b) Bend waveguide with an adjacent straight waveguide to form the coupling region.

Based on the aforementioned analysis, a gap of 500 nm and a bent waveguide with a bend radius of 20 μm are utilized in the two racetrack MRRs. The lengths of the straight waveguides

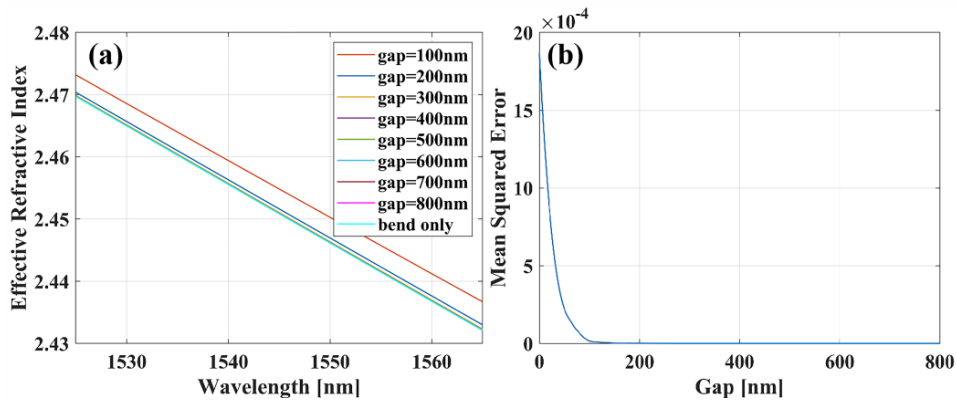


Fig. 5. Effective refractive index simulation results for the two structures in Fig. 4. (a) The n_{eff} at different wavelengths. (b) MSE of the n_{eff} for 10 equally spaced wavelengths in the C-band between the two structures.

are 140 μm and 190 μm , respectively, resulting in cavity perimeters of $\sim 405.66 \mu\text{m}$ and $\sim 505.66 \mu\text{m}$. Figure 6 depicts the microscope image of the two MRRs. The spectra are shown in Figs. 7(a) and (b), where the free spectral ranges (FSRs) are approximately 1.5 nm and 1.2 nm for the short and long cavities, respectively.

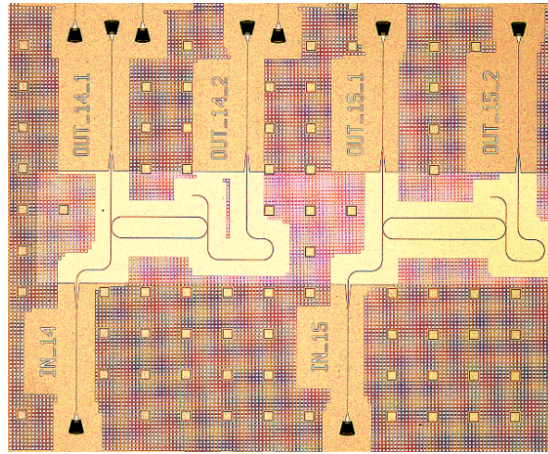


Fig. 6. Microscope image of two MRRs with the bent waveguide radius of 20 μm and straight waveguide lengths being 140 μm and 190 μm , respectively.

The previous analysis shows that the effective refractive index of the waveguide can be determined from the spectra of two racetrack MRRs. The spectra of the MRRs can be calculated based on a given $n_{\text{eff}}(\lambda)$ in a cubic polynomial and then compared to the test spectra to verify the accuracy of the given $n_{\text{eff}}(\lambda)$. However, there are potentially infinite values for the four coefficients of $n_{\text{eff}}(\lambda)$. It is a very time-consuming process to go through all of them to find the optimal one. Therefore, we decided to use an optimization algorithm to speed up the search process.

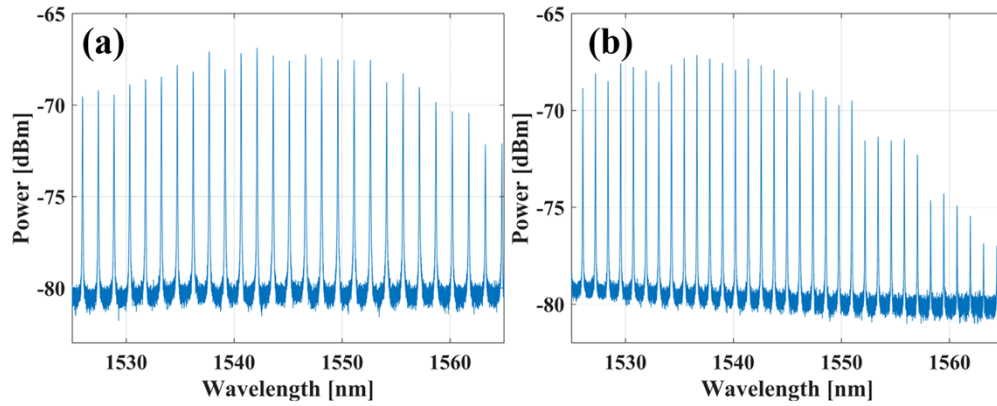


Fig. 7. The measured spectra of two MRRs with straight waveguide length being (a) 140 μm , and (b) 190 μm , respectively.

3. Extraction of effective refractive index and group index

We use the characteristic particle swarm optimization (CPSO) algorithm [28] to find the four coefficients for the cubic polynomial function $n_{\text{eff}}(\lambda)$. The CPSO has the ability to escape local extremes and optimize more efficiently. The iterative formula for this algorithm is

$$\begin{aligned}\vec{v} &= \omega \cdot \vec{v} + c_1 \cdot r_1 \cdot (\vec{p}_{\text{best}} - \vec{x}) + R \cdot \vec{r}, \\ \vec{x} &= \alpha \cdot \vec{g}_{\text{best}} + (1 - \alpha) \cdot \vec{x} + \vec{v}, \\ \alpha &= (i - 1)/(N - 1).\end{aligned}\tag{11}$$

where $\vec{v}$ and $\vec{x}$ are the velocity and position of the particle, respectively, $\vec{p}_{\text{best}}$ is the optimal position of the individual, $\vec{g}_{\text{best}}$ is the optimal position of the group, ω is the inertia constant, c_1 is the personal weight, R is a random factor with arbitrarily chosen value, r_1 is a random number from 0 to 1, $\vec{r}$ is a random vector in which each dimension is a random number from -1 to 1 , α is the global best weight, N is the total number of particles in the population, and i is the serial number of the particle ($i = 1, 2, \dots, N$). The CPSO algorithm follows the same process as the traditional PSO algorithm. The individual and population optimums are continuously updated according to the results of the calculations during the iterative process.

The current optimization problem is to find suitable $n_{\text{eff}}(\lambda)$ so that the calculated spectra can match the measured spectra as much as possible. The position $\vec{x}$ for one particle corresponds to a possible solution for $n_{\text{eff}}(\lambda)$. A simple method is to assign the four coefficients in Eq. (2) to each dimension of the particle's position in the algorithm. However, this approach poses a challenge in defining the range of each dimension. If the range is too narrow, the algorithm may not be able to find the optimal value; whereas, if the range is too wide, it may result in a lengthy convergence time. Fortunately, the foundry provides the error range of the waveguide dimensions in advance, enabling us to determine the range of variation for the effective refractive index. Therefore, we can define each dimension of the particle's position in terms of the effective refractive index at a certain wavelength. This allows us to establish the range of the particle's motion.

Based on the fab data, the width of the waveguide varies within a range of ± 50 nm, while the height and slab thickness vary within a range of ± 10 nm. In addition, the angle of the sidewalls ranges from 75 to 85 degrees. The nominal waveguide cross section and the maximum and minimum variations are shown in Fig. 8(a). The corresponding effective refractive indices for the three cross sections in Fig. 8(a) are shown in Fig. 8(b).

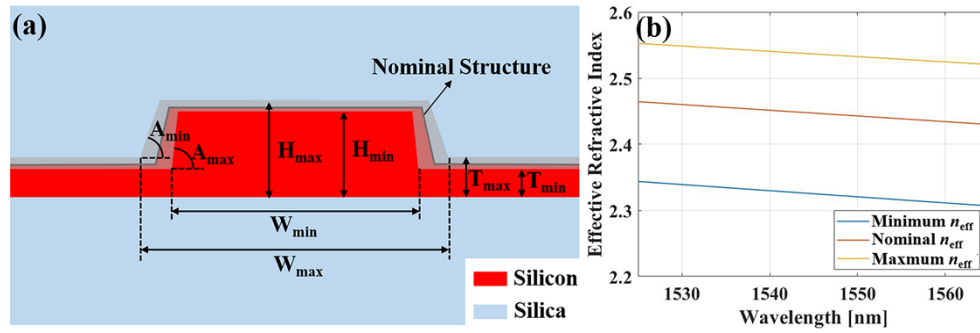


Fig. 8. (a) Waveguide geometry where process errors lead to changes in waveguide shape. (b) The effective refractive index of the nominal waveguide geometry and the maximum and minimum effective refractive index under changes in the waveguide geometry.

As the effective refractive index is related to the wavelength by a cubic polynomial, only four sets of wavelength and corresponding effective refractive index data are required to determine the coefficients of the polynomial. To more finely tune the four polynomial coefficients, we selected 20 wavelengths uniformly in the C-band and used their n_{eff} as the optimization target. In other words, the dimension of the CPSO we use is 20. To evaluate the degree of match between the computed and measured spectra, we used the sum of resonance wavelength location difference between their respective peaks as the algorithm's FOM.

All results with FOM less than the average full width at half maxima (FWHM) of all resonance peaks are considered as possible correct results, due to the spectrometer's inability to accurately determine peak wavelength during test and the possibility of jitter in the spectra. We use the CPSO algorithm to perform a million iterations and record the data that meets the above conditions. It is important to note that, as shown in Eq. (11), the velocity term $\vec{v}$ contains a random number ($\vec{r}$) with constant amplitude (R). This means that even if the algorithm has found the global optimum, the velocity will not be 0 and will continue to perform the search. Therefore, a large number of iterations is necessary not only to find the true global optimum but also to search for enough n_{eff} to meet the requirement. All the n_{eff} satisfying the aforementioned condition is shown in Fig. 9.

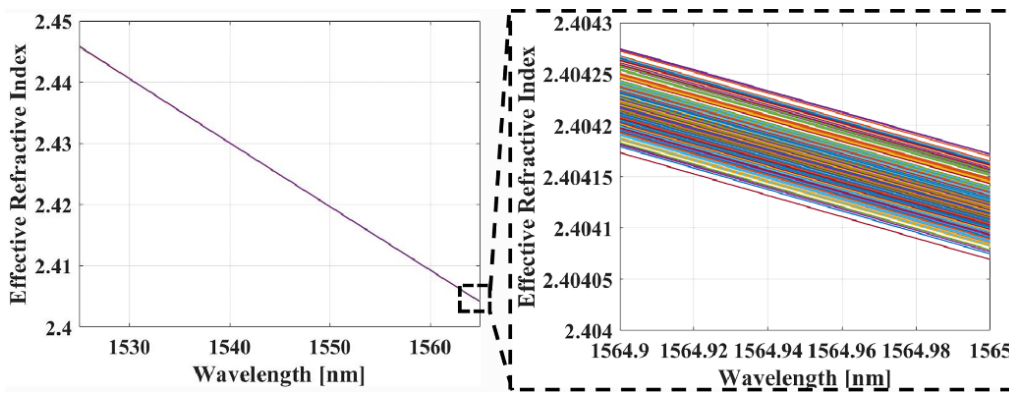


Fig. 9. The effective refractive indices recorded by the CPSO along one million iterations.

Figure 9 shows that the n_{eff} meeting the criteria are very close to each other, but their group indexes are quite different, as shown in Fig. 10(a). Therefore, to choose the most reasonable effective refractive index, we used the following equation to calculate the group index at the

resonance wavelengths in the measured spectrum.

$$FSR = \frac{\lambda^2}{n_g L} \quad (12)$$

where L is the circumference of the MRR. The effective refractive index is determined by comparing the n_g given by Eq. (12) using measured spectra with those in Fig. 10(a) and selecting the result with the smallest MSE, as shown in Fig. 10(b). The spectra calculated from this n_{eff} are compared with the measured spectra, as shown in Fig. 11, which shows that the simulated and measured spectra match well.

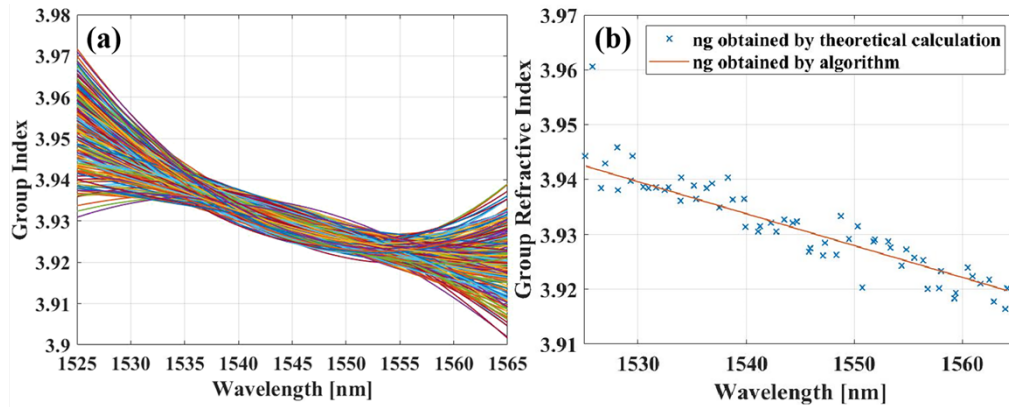


Fig. 10. (a) The group indices corresponding to n_{eff} in Fig. 7. (b) Group index obtained from algorithmic calculations and measured spectra.

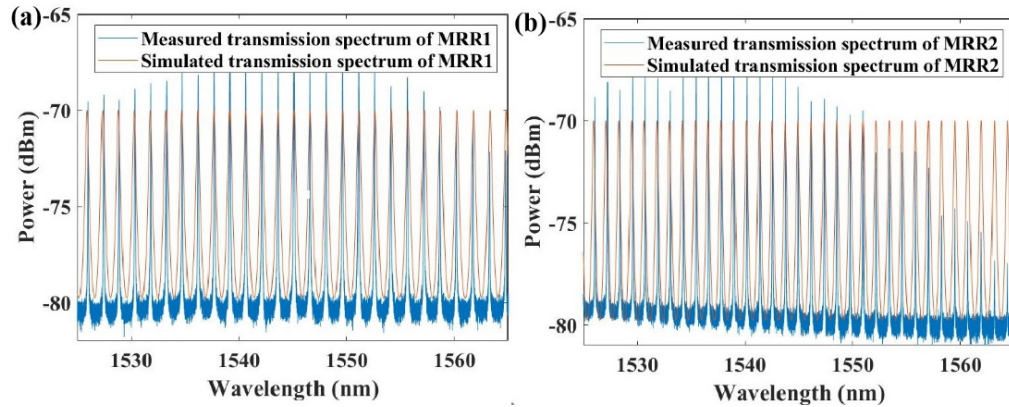


Fig. 11. The measured and calculated spectra of two MRRs with straight waveguide length being (a) 140 μm (b) 190 μm, respectively.

4. Waveguide geometric extraction

The cross section of the rib waveguide is described by W , H , T , and A , which are the width, height, slab thickness, and sidewall angle, respectively. Changing these parameters will result in a corresponding change in the effective refractive index. Therefore, a global scan using the CPSO can be conducted to identify the waveguide cross section that best matches the effective refractive

index obtained from section 3. It should be noted that scanning the four parameters using the particle swarm algorithm is a relatively simple process. The optimization of 800 parameters has been successfully achieved in [28].

To further verify that the identified waveguide structure was the only one available, we first performed a search from the starting cross section with the smallest area and then a second search from the starting cross section with the largest area. The results of the two searches are shown in Table 1. The final result retains the first decimal place, and the W , H , T , A of the final waveguide cross section are 461 nm, 197.4 nm, 68.2 nm, and 75.6° , respectively. The algorithm consistently identified the same optimal structure in both searches. Figure 12 shows the target $n_{\text{eff}}(\lambda)$ and the effective refractive index of the waveguide cross section that the algorithm eventually finds to best fit that target $n_{\text{eff}}(\lambda)$.

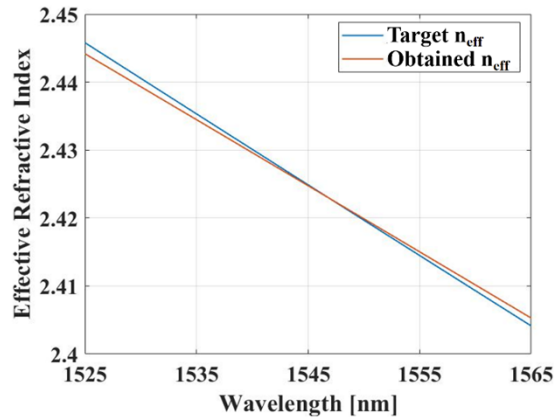


Fig. 12. The target effective refractive index and the effective refractive index obtained by the algorithm searching the waveguide cross section.

Table 1. Waveguide cross section obtained from two searches

	W (nm)	H (nm)	T (nm)	A
First search	460.988	197.442	68.189	75.632
Second search	461.023	197.395	68.212	75.601

To confirm the accuracy of the waveguide geometry obtained, we dice the chip and use the STEM to obtain the actual waveguide cross section. The resolution of the STEM is 0.2 nm, which is sufficient to provide accurate information on the dimensions of the waveguide cross sections. Figure 13 shows the STEM result with the calculated result using the CPSO. It can be seen from the figure that the deviation of W , H , T , A are 4.4 nm, 1.9 nm, 0.1 nm and 0.8° , respectively. This validates the effectiveness of our method in predicting the structural parameters of silicon rib waveguides.

It should be noted that, despite the use of a curved waveguide with a radius of 20 μm to minimize the difference between the bent and straight waveguide, a slight difference remains between their n_{eff} and n_g . We calculated the mean value of their n_{eff} difference in the C-band to be 2.6×10^{-5} based on the results in Fig. 3(a), and we also calculated the mean value of their n_g difference in the C-band to be 3.5×10^{-5} . Taking the initial waveguide cross-sectional dimensions as a reference (W of 450 nm, H of 200 nm, T of 70 nm, A of 80 degrees), we simulated the changes in n_{eff} and n_g when the waveguide has the dimensions given by the STEM measurement. In the C-band, the average discrepancy in n_{eff} is 1.16×10^{-2} , which is much larger than the difference in n_{eff} between straight and bent waveguides. Similarly, the average discrepancy in n_g

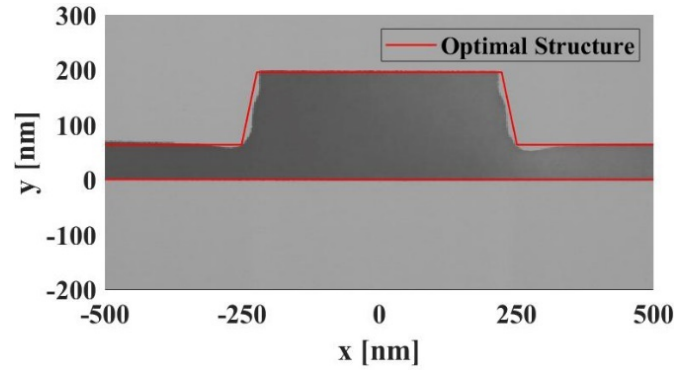


Fig. 13. The actual waveguide cross section measured using the STEM and the optimal waveguide cross section found by the CPSO.

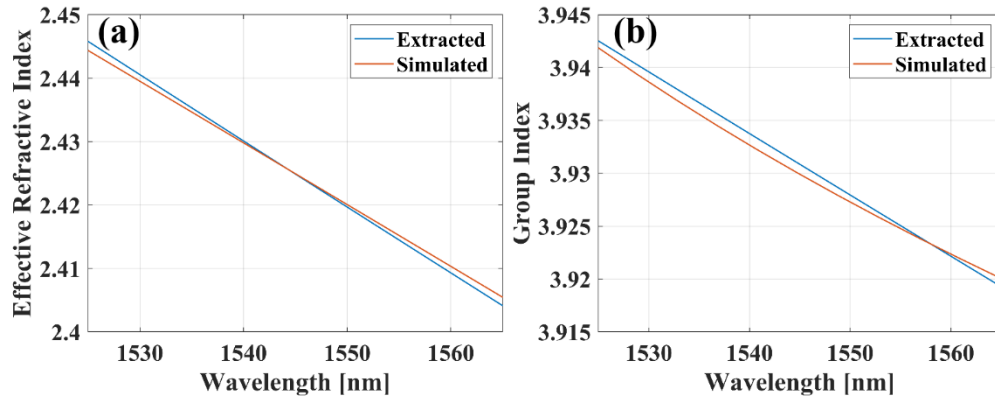


Fig. 14. Comparison of (a) n_{eff} and (b) n_g obtained from simulations using the STEM cross section with the extracted results using the MRR spectra and cubic polynomial fitting.

is 7.2×10^{-3} , which is also significantly larger than the difference in n_g between straight and bent waveguides.

Furthermore, the effective and group refractive indices of the cross section of the waveguide obtained by STEM have been simulated and compared with the results obtained by our algorithm, as illustrated in Fig. 14. It can be observed that the n_{eff} and n_g extracted by our algorithm are in close agreement with the values obtained from simulations using the STEM cross section. The maximum error for the effective refractive index is 0.0014, while the maximum error for the group refractive index is 0.0018. A comparison of our results with others in literature is provided in Table 2, which demonstrates that our method not only exhibits high extraction accuracy but also has the capacity to extract the maximum number of waveguide geometrical parameters.

Table 2. Comparison with other waveguide characterization works

Ref.	Material	Structure	Δn_{eff}	Δn_g	Number of extracted geometry parameters
[18]	Si	MZI	0.002	0.0042	/
[19]	Si	MZI	1.1×10^{-5}	0.001	2
[26]	InP	Waveguide crossing	/	0.01	/
[27]	InP	Point reflector optical waveguide	/	0.003	1
This work	Si	MRR	0.0014	0.0018	4

5. Conclusion

A methodology for extracting the effective and group refractive indices (n_{eff} and n_g) of SOI waveguides is presented. Using numerical simulations, we show that the $n_{\text{eff}}(\lambda)$ of typical silicon waveguides can be accurately described by cubic polynomials with λ as an independent variable. We prove theoretically that the four coefficients of such a cubic polynomial can be determined from the transmission spectra of two racetrack MRRs with the same cross section and different perimeters. We demonstrate the $n_{\text{eff}}(\lambda)$ fitting experimentally using two SOI MRRs fabricated with 193 nm lithography. A modified PSO algorithm is used to search for cubic polynomial coefficients. Based on the extracted function $n_{\text{eff}}(\lambda)$, we estimate the cross-sectional dimension of the SOI ridge waveguide using the FDM for mode solving and the same PSO algorithm for structural parameter estimation. The estimated cross-sectional dimension is in agreement with the results obtained by STEM. The proposed method provides a straightforward way for n_{eff} and n_g extraction and a non-invasive way for cross-sectional dimension estimation. This method can be extended to other material systems and structures, which is significant for waveguide characterization and fabrication process evaluation in various platforms.

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Data availability. Data underlying the results presented in this paper are not publicly available at this time but may be obtained from the authors upon reasonable request.

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